

22 Physical solution models

22.1 Concept of nearest-neighbour bond energies

The modelling of solution phases described in Chapters 20 and 21 was based on the proper expression for the ideal entropy of mixing assuming random mixing within each sublattice. The rest of the modelling was purely mathematical and was not related directly to any physical effects. The present chapter is devoted to models based on more physical considerations. In particular, the interaction energies between atoms will be considered. A very simple and useful way of modelling the thermodynamic properties of a binary solution is based upon the assumption that the energy of the whole system is the sum of the bond energies between neighbouring atoms. In the simplest case one only considers the energies of pairs of nearest neighbours. The formation of the solution from the pure components can then be regarded as a chemical reaction between different kinds of bonds, similar to a reaction between molecules



We have here chosen to distinguish between an A–B bond and a B–A bond although they are of course quite equivalent if all the lattice sites are equivalent.

The reaction gives a change of energy which we may denote by 2ν and regard as an **exchange energy**. Since our aim is to construct an expression for the Gibbs energy of the solution we shall consider the Gibbs energies of the bonds rather than the internal energies. This actually means that we take into account not only the bond energies but also their temperature dependence.

$$2\nu = g_{AB} + g_{BA} - g_{AA} - g_{BB}. \quad (22.2)$$

In some cases g_{AB} and g_{BA} may be different and it may thus be useful to define two different quantities from the beginning,

$$\nu_{AB} = g_{AB} - g_{AA}/2 - g_{BB}/2 \quad (22.3)$$

$$\nu_{BA} = g_{BA} - g_{AA}/2 - g_{BB}/2, \quad (22.4)$$

where $\nu_{AB} + \nu_{BA} = 2\nu$. All bonds of each kind are assumed to have the same energy independent of the local composition. The Gibbs energy contribution from the bond energies can thus be evaluated by counting the number of bonds, N_{AA} , N_{BB} , N_{AB} and N_{BA} .

$$\Delta G = \sum_i \sum_j N_{ij} g_{ij}. \quad (22.5)$$

This is the mathematical definition of the nearest-neighbour bond energy model. In the next section we shall evaluate N_{ij} and add the contribution due to the entropy of configurational disorder.

The counting of the number of bonds of each kind can be done with different degrees of ambition. In the simplest treatment, which is called the Bragg–Williams model, one assumes that the atoms are placed at random on the sites in the crystal and it leads to an expression which is identical to the so-called regular solution model. It may thus be used to justify the regular solution model. In more ambitious treatments one tries to calculate how the ν value influences the number of bonds. A positive ν value indicates that the A and B atoms do not like to mix with each other and, if they have been mixed with each other in a solution, they should at least try to arrange themselves in such a way that there are less A–B bonds than in a random arrangement. A negative ν value, on the other hand, would favour arrangements where the A atoms are surrounded by more B atoms than in a random arrangement. Such effects will be considered later in this chapter using an approximation called the quasi-chemical approach. It is primarily based on a random mixture of the nearest neighbour bonds. In Kikuchi's cluster variation method one considers the random mixture of larger clusters. In principle, one should get an exact description of the configurational entropy by going to clusters of infinite size but that is not practically possible, nor is it necessary. A sufficiently good result is probably obtained by including just a few cluster sizes. It is interesting to note that in the cluster variation method one estimates the energy of a cluster as a sum of its bond energies (also called pair energies), assuming that each kind of pair energy is a constant, independent of the local and global composition.

The concept of nearest neighbour bond energies is closely related to the concept of molecules with a Gibbs energy of formation for each kind of molecule but it is much more difficult to justify. In a substance with molecules the atoms are actually present as groups of atoms bound together tightly and it is often a good approximation to neglect interactions between atoms in different molecules. However, the splitting up of the total energy of a crystal into a large number of bond energies is quite arbitrary and one may, for instance, choose to consider or neglect next-nearest neighbour bonds and to consider bond energies related to pairs of atoms or to larger groups of atoms, i.e. clusters. Even if one decides to consider only pairs of atoms or larger groups of atoms, the energy of the different kinds of bonds is rather arbitrary unless one has information relating to different types of ordering. This being so, it is doubtful whether a rather random distribution of atoms can be described with cluster energies evaluated from ordered arrangements. A very crude but useful way of improving the pair energy model would be to assume that only part of the excess Gibbs energy in a disordered state is of such short-range character that it can affect short- and long-range order. That approach would give an additional adjustable parameter to be used in the description of thermodynamic and configurational information.

The justification of the nearest-neighbour bond energy model has to come from its success in representing experimental facts. It has been found very useful in giving qualitative explanations of many phenomena in alloy systems but less successful in accounting for experimental data in detail. There are many modifications of the basic treatment but we

shall first consider the simplest possible approach, the random mixing model of Bragg and Williams.

Exercise 22.1

Suppose one has found experimentally that ν is a constant across a binary system. This result may be interpreted by assuming that all the bond energies are independent of composition. However, suppose one has some theoretical reason to expect that g_{AA} and g_{BB} should vary linearly across the system. Would it then be possible to explain the same experimental result?

Hint

Let $g_{AA} = g_{AA}^0 + ax_B$ and $g_{BB} = g_{BB}^0 + bx_B$.

Solution

$2\nu = g_{AB} + g_{BA} - g_{AA}^0 - ax_B - g_{BB}^0 - bx_B$. Yes, it is possible to eliminate x_B if $g_{AB} + g_{BA} - ax_B - bx_B = 0$ but there are no good reasons to expect such a relation.

22.2 Random mixing model for a substitutional solution

A solid phase where atoms of different components can substitute for each other, i.e. occupy the same kind of lattice sites, is called a substitutional solution. The composition of such a solution is conveniently described with the molar contents of the atoms, x_A and x_B in a binary solution. In order to describe the arrangement of the atoms relative to each other, it may be convenient to introduce the fractions of bonds, p_{AA} , p_{AB} , p_{BA} and p_{BB} . The notation p is chosen because the fraction of a certain kind of bond is equal to the probability of finding that kind of bond. Of course, $\sum p_{ij} = 1$. The fractions are also related through the composition and for the simple case where all the atoms have the same number of bonds this condition can be written in two ways

$$x_A = p_{AA} + (p_{AB} + p_{BA})/2 \quad (22.6)$$

$$x_B = p_{BB} + (p_{AB} + p_{BA})/2. \quad (22.7)$$

In the case of random mixing we get

$$p_{AA} = x_A^2 \quad (22.8)$$

$$p_{AB} = x_A x_B = p_{BA} \quad (22.9)$$

$$p_{BB} = x_B^2. \quad (22.10)$$

Let us assume that all atoms have the same number of nearest neighbours. We can thus introduce a single z value as the coordination number. The total number of bonds in a system containing one mole of atoms is thus $zN^A/2$ since each bond is shared

between two atoms. N^A is Avogadro's number. One will thus obtain, for instance, $N_{AA} = p_{AA} \cdot zN^A/2$ and for one mole of pure A we obtain

$$^{\circ}G_A = g_{AA} \cdot 1 \cdot zN^A/2. \quad (22.11)$$

The Gibbs energy contribution from all the bond energies is

$$\Delta G_m = \sum_i \sum_j N_{ij} g_{ij} = (p_{AA}g_{AA} + p_{AB}g_{AB} + p_{BA}g_{BA} + p_{BB}g_{BB}) \cdot zN^A/2. \quad (22.12)$$

By inserting v_{AB} and v_{BA} we obtain

$$\begin{aligned} \Delta G_m &= [p_{AB}v_{AB} + p_{BA}v_{BA} + g_{AA}(p_{AA} + p_{AB}/2 + p_{BA}/2) \\ &\quad + g_{BB}(p_{BB} + p_{AB}/2 + p_{BA}/2)] \cdot zN^A/2 \\ &= x_A ^{\circ}G_A + x_B ^{\circ}G_B + [p_{AB}v_{AB} + p_{BA}v_{BA}] \cdot zN^A/2. \end{aligned} \quad (22.13)$$

We shall now add the contribution due to the entropy of mixing by considering random mixing of atoms. We get by substituting $x_A x_B$ for p_{AB} and p_{BA} and inserting $2v = v_{AB} + v_{BA}$,

$$G_m = x_A ^{\circ}G_A + x_B ^{\circ}G_B + v z N^A x_A x_B + RT(x_A \ln x_A + \ln x_B). \quad (22.14)$$

Exercise 22.2

Compare the final G_m expression for this random mixing model with the corresponding expression according to the regular solution model.

Hint

According to Section 20.4, the regular solution model gives $^E G_m = x_A x_B {}^0L$.

Solution

The random bond energy model yields $^E G_m = v z N^A x_A x_B$. The two models are exactly related to each other by ${}^0L = v z N^A$.

22.3 Deviation from random distribution

In order to describe non-random solutions we shall make extensive use of the bond probabilities, p_{ij} . In order to describe both long- and short-range order we need two independent internal variables and they can be defined in many ways. A convenient choice is the following,

$$K = (p_{AB} + p_{BA})/2 \quad (22.15)$$

$$L = (p_{AB} - p_{BA})/2, \quad (22.16)$$

and it yields

$$p_{AB} = K + L; \quad dp_{AB} = dK + dL \quad (22.17)$$

$$p_{BA} = K - L; \quad dp_{BA} = dK - dL \quad (22.18)$$

$$p_{AA} = x_A - (p_{AB} + p_{BA})/2 = x_A - K; \quad dp_{AA} = dx_A - dK \quad (22.19)$$

$$p_{BB} = x_B - (p_{AB} + p_{BA})/2 = x_B - K; \quad dp_{BB} = -dx_A - dK. \quad (22.20)$$

Long-range order can only be described with the use of two or more sublattices. The situation in an ordered alloy with two sublattices can be described with the site fractions which, in turn, can be expressed through the bond probabilities. Assuming that all the bonds go between atoms in two different sublattices we find

$$y'_A = p_{AA} + p_{AB} = x_A + L; \quad dy'_A = dx_A + dL \quad (22.21)$$

$$y'_B = p_{BB} + p_{BA} = x_B - L; \quad dy'_B = -dx_A - dL \quad (22.22)$$

$$y''_A = p_{AA} + p_{BA} = x_A - L; \quad dy''_A = dx_A - dL \quad (22.23)$$

$$y''_B = p_{BB} + p_{AB} = x_B + L; \quad dy''_B = -dx_A + dL. \quad (22.24)$$

Long-range order means that element A prefers one sublattice and B the other. It is conveniently defined as

$$l.r.o. = y'_A - y''_A = p_{AA} + p_{AB} - p_{AA} - p_{BA} = 2L. \quad (22.25)$$

Short-range order means that the atoms with given site fractions do not arrange themselves at random within each sublattice. Random distribution would yield the following probabilities of finding various bonds between the two sublattices and we shall still assume that there are no bonds within a sublattice.

$$p_{AA} = y'_A y''_A \quad (22.26)$$

$$p_{AB} = y'_A y''_B \quad (22.27)$$

$$p_{BA} = y'_B y''_A \quad (22.28)$$

$$p_{BB} = y'_B y''_B. \quad (22.29)$$

Short-range order may be defined as the deviation from this arrangement

$$\begin{aligned} s.r.o. &= [p_{AB} - y'_A y''_B + p_{BA} - y'_B y''_A]/2 \\ &= [2K - (x_A + L)(x_B + L) - (x_B - L)(x_A - L)]/2 \\ &= K - L^2 - x_A x_B. \end{aligned} \quad (22.30)$$

We know from Section 19.8 that a random distribution within each sublattice would yield the following configurational entropy for 1 mole of atoms,

$$S_m/R = -(y'_A \ln y'_A + y'_B \ln y'_B + y''_A \ln y''_A + y''_B \ln y''_B). \quad (22.31)$$

This does not account for short-range order. In an attempt to treat that case, one could start by considering a random distribution of the bonds,

$$S_m/R = -(z/2)(p_{AA} \ln p_{AA} + p_{AB} \ln p_{AB} + p_{BA} \ln p_{BA} + p_{BB} \ln p_{BB}). \quad (22.32)$$

However, this does not reduce to the previous expression when the random values for the four p_{ij} are inserted. That condition can be satisfied if we divide each p_{ij} under an $\ln$ sign by its random value, which will make the whole expression go to zero for a random case, and then add the previous expression, which should be the correct one for the random case and a good approximation for small deviations from randomness.

$$\begin{aligned}
 -S_m/R &= (z/2)[p_{AA} \ln(p_{AA}/y'_A y''_A) + p_{AB} \ln(p_{AB}/y'_A y''_B) \\
 &\quad + p_{BA} \ln(p_{BA}/y'_B y''_A) + p_{BB} \ln(p_{BB}/y'_B y''_B)] \\
 &\quad + [y'_A \ln y'_A + y'_B \ln y'_B + y''_A \ln y''_A + y''_B \ln y''_B]/2. \quad (22.33)
 \end{aligned}$$

The contribution from the bond energies was given by Eq. (22.13). The complete expression will thus be

$$\begin{aligned}
 G_m &= x_A^\circ G_A + x_B^\circ G_B + [p_{AB} v_{AB} + p_{BA} v_{BA}] \cdot z N^A/2 \\
 &\quad + RT(z/2)[(p_{AA} \ln(p_{AA}/y'_A y''_A) + p_{AB} \ln(p_{AB}/y'_A y''_B) \\
 &\quad + p_{BA} \ln(p_{BA}/y'_B y''_A) + p_{BB} \ln(p_{BB}/y'_B y''_B)] \\
 &\quad + RT[y'_A \ln y'_A + y'_B \ln y'_B + y''_A \ln y''_A + y''_B \ln y''_B]/2. \quad (22.34)
 \end{aligned}$$

We shall now apply this general expression to several special cases. However, it should be remembered that the expression is valid only under the assumption that there are two sublattices, that they contain the same number of bonds and that all bonds go between atoms in different sublattices. The simple bcc structure (A2) can order in this way, yielding B2.

Exercise 22.3

Demonstrate that the general equation for ordering reduces to the model for random mixing in a substitutional solution without short-range order.

Hint

In a substitutional solution without long-range order and thus $y'_A = y''_A = x_A$ and $y'_B = y''_B = x_B$ and without short-range order $p_{ij} = y'_i y''_j = x_A x_B$. Furthermore, v_{AB} and v_{BA} must be equal in order to prevent long-range order.

Solution

The first part of the entropy contribution in Eq. (22.34) reduces to zero and the second one to $RT(x_A \ln x_A + x_B \ln x_B)$. In the energy part $p_{AB} v_{AB} + p_{BA} v_{BA}$ reduces to $2x_A x_B v$. Thus, $\Delta G_m = x_A^\circ G_A + x_B^\circ G_B + z N^A x_A x_B + RT(x_A \ln x_A + x_B \ln x_B)$.

22.4 Short-range order

At a high enough temperature long-range order will disappear if $v_{AB} = v_{BA} = v$ and only short-range order remains. For this case, $y'_A = y''_A = x_A$ and $y'_B = y''_B = x_B$ would yield $L = 0$ and $p_{AB} = p_{BA}$. We obtain from Eq. (22.34)

$$\begin{aligned} G_m = & x_A^0 G_A + x_B^0 G_B + v z N^A p_{AB} + RT(z/2) [p_{AA} \ln(p_{AA}/x_A^2) \\ & + 2p_{AB} \ln(p_{AB}/x_A x_B) + p_{BB} \ln(p_{BB}/x_B^2)] \\ & + RT[x_A \ln x_A + x_B \ln x_B]. \end{aligned} \quad (22.35)$$

We have only one internal variable $K = (p_{AB} + p_{BA})/2 = p_{AB}$. Its equilibrium value under constant composition is obtained from

$$\begin{aligned} (\partial G_m / \partial K)_{L, x_A} = & v z N^A + RT(z/2) [-\ln(p_{AA}/x_A^2) - p_{AA}/p_{AA} + 2 \ln(p_{AB}/x_A x_B) \\ & + 2p_{AB}/p_{AB} - \ln(p_{BB}/x_B^2) - p_{BB}/p_{BB}] \\ = & v z N^A + RT(z/2) \ln(p_{AB}^2/p_{AA} p_{BB}) = 0 \end{aligned} \quad (22.36)$$

$$\frac{p_{AB}^2}{p_{AA} p_{BB}} = \exp(-2v/kT). \quad (22.37)$$

This resembles the law of mass action for a chemical reaction between molecules and this method of correcting the entropy expression is thus called the quasi-chemical method. It should be emphasized that p_{AB} is here defined as half the number of AB bonds because the other half is counted as p_{BA} .

It is worth noting that the quasi-chemical method of correcting the entropy expression is valid only for small deviations from randomness, i.e. for low values of v/kT . It is immediately evident that very large values of v/kT will produce unreasonable results. An infinite value of v/kT will make $p_{AB} = p_{BA} = 0$ and thus $p_{AA} = x_A$ and $p_{BB} = x_B$. This implies a separation of the system into two parts, one containing all the A atoms and the other all the B atoms. The configurational entropy should thus be zero but the entropy part of Eq. (22.35) would yield

$$\begin{aligned} S_m = & -R(z/2)[x_A \ln(1/x_A) + 0 + x_B \ln(1/x_B)] - R[x_A \ln x_A + x_B \ln x_B] \\ = & (z/2 - 1)R[x_A + x_B \ln x_B]. \end{aligned} \quad (22.38)$$

This yields the correct value (zero) for $z = 2$, which applies when all the atoms are arranged in a string (the Ising model). For all realistic z values (e.g. $z = 8$ for bcc and $z = 12$ for fcc) the entropy expression yields large negative values. This result emphasizes that the quasi-chemical method should be used only for low values of v/kT , i.e. for low deviations from the ideal entropy. There one can simplify the expression by series expansions,

$$\exp \varepsilon = 1 + \varepsilon + \varepsilon^2/2 \quad (22.39)$$

$$\sqrt{1 + \varepsilon} = 1 + \varepsilon/2 - \varepsilon^2/8 + \varepsilon^3/16, \quad (22.40)$$

yielding

$$p_{AB} = p_{BA} = x_A x_B [1 - 2x_A x_B v/kT - 2x_A x_B (x_A - x_B)^2 (v/kT)^2]. \quad (22.41)$$

$$G_m = x_A {}^0G_A + x_B {}^0G_B + RT(x_A \ln x_A + x_B \ln x_B) + v z N^A x_A x_B [1 - x_A x_B v/kT - (2/3)x_A x_B (x_A - x_B)^2 (v/kT)^2]. \quad (22.42)$$

It is self-evident that the Gibbs energy will decrease due to short-range order, otherwise it would not form. The presence of short-range order will thus stabilize the disordered state and depress the temperature for the transition to a state with long-range order. This conclusion holds independent of the sign of v . Positive v will result in a miscibility gap at low temperatures if that reaction is not prevented by other reactions. Already above the miscibility gap a positive v will favour A–A bonds and B–B bonds and result in clusters. This tendency grows very strong as the consolute point is approached on cooling of a system with the correct composition. The tendency to actually separate into two phases will thus decrease and the consolute point of the miscibility gap will be depressed to lower temperatures. Also, the consolute point will be flatter than calculated from the regular solution model because the effect will be strongest close to the consolute point. This effect is quite noticeable in the liquid state where the presence of clusters may give rise to opalescence close to the consolute point. In the solid case the effect is counteracted by coherency stresses due to the difference in atomic sizes.

In order to treat this effect properly it is not enough to consider nearest neighbours. It is not even enough to extend the consideration to larger clusters. It has to be treated with a mathematical technique called the renormalization group approach. The resulting shape of the miscibility gap is non-analytical, especially close to the maximum.

Exercise 22.4

In the next section we shall find that long-range order is predicted to occur below $T_{tr} = 2x_A x_B (-vz/k)$ if the effect of short-range order is neglected. Evaluate the short-range order in a 50/50 bcc alloy at this temperature in order to test if the neglect of short-range order is serious.

Solution

$p_{AA} p_{BB} = (p_{AB})^2 \exp(2v/kT_{tr}) = (p_{AB})^2 \exp[2v/2x_A x_B (-vz)] = (p_{AB})^2 \exp(-4/z)$; Insert p_{AA} and p_{BB} from Eqs (22.6) and (22.7): $(0.5 - p_{AB})^2 = 0.606(p_{AB})^2$; $p_{AB} = 0.2811$; s.r.o. = $K - L^2 - x_A x_B = 0.2811 - 0 - 0.25 = 0.0311$. This is not negligible. The value of T_{tr} given above may be regarded as a rough estimate but short-range order makes the disordered state more stable and it should depress T_{tr} .

22.5 Long-range order

Let us again consider negative exchange energies. There is always a tendency for short-range order but as an introduction to the more general case it may be illustrative first to consider long-range order and neglect short-range order which is done by taking all $p_{ij} = y'_i y''_j$. The general equation then only contains y_i variables. The first part of the entropy contribution reduces to zero and with $v_{AB} + v_{BA} = 2v$ we find

$$G_m = x_A {}^\circ G_A + x_B {}^\circ G_B + (y'_A y''_B + y'_B y''_A) \cdot v z N^A / 2 \\ + RT[y'_A \ln y'_A + y'_B \ln y'_B + y''_A \ln y''_A + y''_B \ln y''_B] / 2. \quad (22.43)$$

With a fixed composition there is only one independent internal variable and we may choose any one of the four y_i or the long-range order parameter L . Using the relations between L and the y variables in Eqs (22.19) to (22.24) we thus find the equilibrium from

$$(\partial G_m / \partial L)_{x_A} = (y''_B + y'_A - y''_A - y'_B) \cdot v z N^A / 2 + RT[y'_A / y'_A + \ln y'_A \\ - y'_B / y'_B - \ln y'_B - y''_A / y''_A - \ln y''_A + y''_B / y''_B + \ln y''_B] / 2 = 0 \quad (22.44)$$

$$\frac{kT}{-vz} = \frac{y'_A + y'_B - y''_A - y''_B}{\ln(y'_A y''_B / y'_A y''_B)}. \quad (22.45)$$

using $R = kN^A$. It is evident that $y'_A = y''_A = x_A$ (and thus $y'_B = y''_B = x_B$) is a solution for all T values and it represents a completely random distribution. However, below a particular T value another kind of solution appears and it represents long-range order. That temperature has to be calculated as the limiting value of T where y'_A and y''_A of the new kind of solution approach x_A since the numerator and denominator are both zero. It is obtained by taking the ratio of their derivatives,

$$\frac{kT_{tr}}{-vz} = \frac{1 + 1 + 1 + 1}{1/y'_A + 1/y''_B + 1/y''_A + 1/y'_B} = \frac{4}{2/x_A + 2/x_B} = 2x_A x_B. \quad (22.46)$$

This relation defines the transition line, i.e. the boundary of the ordering region in the T, x_B phase diagram, see Fig. 21.3(a). It has a parabolic shape just as the spinodal curve for positive v values. Its maximum is found at $x_A = x_B = 0.5$ and $T_{tr}^{\max} = -vz/2k$. The present result is identical to the result of the compound energy formalism in Section 21.5 and the two models are related by $\Delta {}^\circ G_{AB+BA} = vzN^A$. The Bragg–Williams model thus provides a physical interpretation of the model parameter in this simple application of the compound energy formalism.

Above the ordering region the present model predicts complete disorder and the expression for G_m degenerates to

$$G_m = x_A {}^\circ G_A + x_B {}^\circ G_B + vzN^A x_A x_B + RT(x_A \ln x_A + x_B \ln x_B), \quad (22.47)$$

because $y'_A = y''_A = x_A$ and $y'_B = y''_B = x_B$. This expression was derived in Section 22.2 and it was then emphasized that vzN^A is identical to the regular solution parameter 0L . Within the limitations of the nearest-neighbour bond energy model it would thus be possible to predict the ordering behaviour of a binary solution at low temperatures from the value of the regular solution parameter at high temperatures. However, as mentioned

in Section 22.1, it should be realized that part of the energy may be of a more long-range character than nearest-neighbour interactions. It is thus possible that the ordering tendency should be represented by a v value which is not quite equal to $^0L/zN^A$.

Exercise 22.5

We have derived an expression for the transition line for ordering in a binary system, taking into account a gradual increase of long-range order with decreasing temperature. Now, formulate a more primitive theory by considering only two possible states, complete order and complete disorder. Calculate the transition temperature and compare with the result just obtained for the critical temperature. Limit the calculation to a 50/50 alloy.

Hint

One can directly calculate the Gibbs energy contributions from configurational entropy and bond energies for the two states. Do not introduce the exchange energy v until the two states are compared.

Solution

In the disordered state the configurational entropy for 1 mole of atoms is $R\ln 2$ and the energy due to bonds is $(x_A^2 g_{AA} + 2x_A x_B g_{AB} + x_B^2 g_{BB}) \cdot zN^A/2 = (g_{AA} + 2g_{AB} + g_{BB}) \cdot zN^A/8$ for $x_A = x_B = 0.5$. In the ordered state there is no configurational entropy and the bonds give an energy of $g_{AB}zN^A/2$. At equilibrium: $-RT \ln 2 + (g_{AA} + 2g_{AB} + g_{BB})zN^A/8 = g_{AB}zN^A/2$; $RT \ln 2 = (g_{AA} - 2g_{AB} + g_{BB})zN^A/8 = -vzN^A/4$; $T = -vz/4k \ln 2$. This should be compared with $T_{tr} = -vz/2k$. The ratio is $T/T_{tr} = 1/2 \ln 2 = 0.72$.

Exercise 22.6

Apply the condition of stability limit and verify that the transition temperature falls at the limit of stability as it should for a second-order transition.

Hint

The stability condition, Eq. (6.54), can here be written as $(\partial^2 G_m / \partial L^2)_{x_A} = 0$. We have already an expression for $(\partial G_m / \partial L)_{x_A}$.

Solution

$(\partial^2 G_m / \partial L^2)_{x_A} = (1 + 1 + 1 + 1)vzN^A/2 + RT[1/y'_A + 1/y'_B + 1/y''_A + 1/y''_B]/2$. For the disordered state, $y'_A = y''_A = x_A$, we get $(\partial^2 G_m / \partial L^2)_{x_A} = 2vzN^A + RT(1/x_A + 1/x_B) = 2vzN^A + RT/x_A x_B = 0$ yielding $T_{limit} = -2vzx_A x_B/k$ in agreement with the expression for T_{tr} .

22.6 Long- and short-range order

In the general case there will be both long- and short-range order and the situation will be described with two independent internal variables. We shall use K and L and with the relations of the various y and p quantities to L and K , given by Eqs (22.17) to (22.24), we obtain for equilibrium,

$$(\partial G_m / \partial K)_L = (v_{AB} + v_{BA}) \cdot zN^A / 2 + RT(z/2)[- \ln(p_{AA}/y'_A y''_A) - 1 + \ln(p_{AB}/y'_A y''_B) + 1 + \ln(p_{BA}/y'_B y''_A) + 1 - \ln(p_{BB}/y'_B y''_B) - 1] = 0 \quad (22.48)$$

$$\frac{p_{AB} p_{BA}}{p_{AA} p_{BB}} = \exp[-(v_{AB} + v_{BA}) \cdot z / kT] \quad (22.49)$$

$$\begin{aligned} (\partial G_m / \partial L)_K &= (v_{AB} - v_{BA}) \cdot zN^A / 2 + RT(z/2)[-p_{AA}/y'_A + p_{AA}/y''_A \\ &\quad + \ln(p_{AB}/y'_A y''_B) + 1 - p_{AB}/y'_A - p_{AB}/y''_B - \ln(p_{BA}/y'_B y''_A) \\ &\quad - 1 + p_{BA}/y'_B + p_{BA}/y''_A + p_{BB}/y'_B - p_{BB}/y''_B] \\ &\quad + RT[\ln y'_A + 1 - \ln y'_B - 1 - \ln y''_A - 1 + \ln y''_B + 1] / 2 \\ &= (zRT/2) \ln(p_{AB}/p_{BA}) + (zR/2k)(v_{AB} - v_{BA}) - (z - 1) \\ &\quad \times (RT/2) \ln(y'_A y''_B / y'_B y''_A) = 0. \end{aligned} \quad (22.50)$$

This equation was simplified using $p_{AA} = y'_A - p_{AB} = y''_A - p_{BA}$ and $p_{BB} = y'_B - p_{BA} = y''_B - p_{AB}$ from Eqs (22.21) to (22.24). It is easy to make numerical calculations from Eqs (22.49) and (22.50).

In this chapter we have so far discussed ordering of different kinds of atoms. This phenomenon is often called **chemical ordering** or **configurational ordering**. A completely different kind of ordering occurs when a liquid solidifies and the atoms arrange themselves in a regular pattern, the crystalline structure. In this connection the liquid is said to be topologically disordered. The solid $\rightarrow$ liquid reaction can thus be regarded as an order-disorder transition and, evidently, it is a first-order transition. In the first-order transition, illustrated in Fig. 15.4, the superheated, ordered state reaches a stability limit at some high temperature and the undercooled, disordered state reaches a stability limit at some low temperature. The question whether such limits of stability exist for the solid-liquid transition has attracted some attention. The very simple and crude model described in Section 19.3 may be used to model a continuous transition from solid to liquid and it would predict limits of stability. On the other hand, cooling experiments with liquid alloys have revealed that the high entropy of the liquid state, as compared to the solid, disappears on cooling, and extrapolation of the high-temperature data seems to indicate that it approaches the entropy of the solid at some low temperature. In the same range of temperature the viscosity increases drastically and the liquid transforms into a viscous, amorphous state, which has an entropy similar to the solid. It seems that the amorphous state has only a low topological disorder although it is enough to prevent

the topological long-range order found in crystalline solids. It seems that the topological disorder in the amorphous-liquid phase increases gradually with temperature and never shows a transition point. This is somewhat similar to the behaviour of the asymmetric case of chemical ordering illustrated in Fig. 21.4. From the mechanical point of view, one defines the drastic increase of the viscosity as a glass transition. Thermodynamically, one could define a related point where extrapolated data predict that the entropy of liquid and solid should be equal. However, one should not really expect that the entropy of the liquid reaches that of the crystal and then suddenly starts to follow the value of the crystal. Most probably, there is no sharp thermodynamic transition point between the liquid and the amorphous states.

Due to the topological disorder in a liquid, it would not be possible to observe chemical long-range order in liquid alloys. Nevertheless, there are many cases with very strong chemical ordering in liquids, the most typical case is a molten salt where the electric charges make the cations tend to surround themselves with anions and vice versa. A rather realistic model of a molten salt is thus based on the assumption of two sublattices, one for cations and one for anions. That would be to assume complete long-range order and the effect of short-range order could then be added to the model. A problem with such a model is that the coefficients in the chemical formula (the stoichiometric coefficients) will vary with composition if there are cations of different valencies or anions of different valencies.

Another difficulty appears when one wants to model the change in chemical order in a liquid from a high value at a low temperature and to a low value at a high temperature. As demonstrated in Section 22.4 the quasi-chemical approach to short-range order becomes unrealistic at large degrees of short-range order. It has been proposed that this difficulty can be overcome by using $z = 2$, for which the quasi-chemical approach does not break down at high degrees of short-range order. However, that would make the model less physical. Another possibility would be to use a two-sublattice model of the asymmetric type illustrated in Fig. 21.4(a). It can predict a gradual change from very low to very high degrees of order and without a transition point. A further possibility is to mimic the chemical ordering in a liquid by the formation of molecular-like clusters of atoms, so-called associates. Finally, in an attempt to develop the two-sublattice model to a model applicable to many different types of systems and to intermediate cases, the two-sublattice model with complete long-range order has been manipulated in such a way that it can describe high as well as low degrees of ordering.

Exercise 22.7

Calculate the degree of short-range order when the disordered state is just becoming unstable in a second-order transition.

Hint

There are two internal degrees of freedom, K and L . The limit of stability is thus obtained as $g_{KK} \cdot g_{LL} - g_{KL} \cdot g_{LK} = 0$ according to Eq. (6.53). However, L is still zero at that point,

$y'_A = y''_A = x_A$ and $p_{AB} = p_{BA}$. Then it is easily shown that $g_{KL} = g_{LK} = 0$ and $g_{KK} > 0$. The condition is thus $g_{LL} = 0$.

Solution

From Eq. (22.50) we get, using the relations of dL to dp_{ij} and y_i^s from Eqs (22.17) to (22.24): $g_{LL} = (\partial^2 G_m / \partial L^2)_K = (zRT/2)(1/p_{AB} + 1/p_{BA}) - (z-1)(RT/2)(1/y'_A + 1/y''_A + 1/y'_B + 1/y''_B) = 0$; $p_{AB} = x_A x_B z / (z-1) = p_{BA} = K$ and s.r.o. $= K - L^2 - x_A x_B = x_A x_B [z/(z-1) - 1] = x_A x_B / (z-1)$ since $L = 0$.

22.7 The compound energy formalism with short-range order

In Section 22.2 it was assumed that all atoms have the same z value. In Section 22.3 it was further assumed that all bonds go between atoms in two different sublattices. These assumptions were carried over to the treatment of order in Sections 22.3 to 22.6. However, the treatment can be generalized to cases with different coordination numbers, z' and z'' , if it is still assumed that all bonds go between atoms in two different sublattices but that requires that the number of sites, b and c , are related by $bz' = cz''$. The total number of bonds in a system containing one mole of atoms is $bz'N^A = cz''N^A$ if the sum of b and c is made equal to unity. Thus $N_{AA} = p_{AA} \cdot bz'N^A$ and ${}^\circ G_A = g_{AA} \cdot bz'N^A$ and

$$x_A = p_{AA} + bp_{AB} + cp_{BA}; \quad p_{AA} = x_A - bp_{AB} - cp_{BA} \quad (22.51)$$

$$x_B = p_{BB} + bp_{BA} + cp_{AB}; \quad p_{BB} = x_B - bp_{BA} - cp_{AB}. \quad (22.52)$$

Using these expressions for p_{AA} and p_{BB} we obtain, for the Gibbs energy contribution from all the bond energies,

$$\begin{aligned} \Delta G_m &= \sum_i \sum_j N_{ij} g_{ij} = [p_{AA} g_{AA} + p_{AB} g_{AB} + p_{BA} g_{BA} + p_{BB} g_{BB}] \cdot bz'N^A \\ &= [x_A g_{AA} + x_B g_{BB} + p_{AB}(g_{AB} - bg_{AA} - cg_{BB}) \\ &\quad + p_{BA}(g_{BA} - bg_{BB} - cg_{AA})] \cdot z'bN^A/2. \end{aligned} \quad (22.53)$$

We find, by generalizing the definitions of the v quantities,

$$v_{AB} = g_{AB} - bg_{AA} - cg_{BB} \quad (22.54)$$

$$v_{BA} = g_{BA} - bg_{BB} - cg_{AA} \quad (22.55)$$

$$\Delta G_m = x_A {}^\circ G_A + x_B {}^\circ G_B + bz'N^A \cdot [p_{AB}v_{AB} + p_{BA}v_{BA}]. \quad (22.56)$$

The expression for the configurational entropy in Eq. (22.33) will also be modified by replacing $z/2$ by bz' or cz'' and $1/2$ by b or c , yielding for the total Gibbs energy instead of Eq. (22.34),

$$\begin{aligned} G_m &= x_A {}^\circ G_A + x_B {}^\circ G_B + bz'N^A \cdot [p_{AB}v_{AB} + p_{BA}v_{BA}] + RTbz' [p_{AA} \ln(p_{AA}/y'_A y''_A) \\ &\quad + p_{AB} \ln(p_{AB}/y'_A y''_B) + p_{BA} \ln(p_{BA}/y'_B y''_A) + p_{BB} \ln(p_{BB}/y'_B y''_B)] \\ &\quad + RT[b y'_A \ln y'_A + b y'_B \ln y'_B + c y''_A \ln y''_A + c y''_B \ln y''_B]. \end{aligned} \quad (22.57)$$

This may be regarded as a generalization of Eq. (22.34). However, the definitions of v_{AB} and v_{BA} cannot be reconciled with the simple concept of bond energies presented in Section 22.1. The new definition is closely related to the definition of $\Delta^\circ G_{A_bB_c}$ and $\Delta^\circ G_{B_aA_c}$ in the compound energy formalism in Section 21.5, the only difference being the numerical factor $bz'N^A$. However, the compound energy formalism goes one step further. It postulates that phases, with some bonds between atoms in the same sublattice, can be treated with the same formalism. That may not be correct but may be remedied by the use of the I parameters in the excess terms in Section 21.3.

Using Eq. (22.57) one can formally account for the effect of short-range order with the compound energy formalism. The G_m expression in Eq. (21.40) would be modified to

$$\begin{aligned} G_m = & x_A^\circ G_A + x_B^\circ G_B + p_{AB} \Delta^\circ G_{A_bB_c} + p_{BA} \Delta^\circ G_{B_bA_c} \\ & + RTbz' \cdot [p_{AA} \ln(p_{AA}/y'_A y''_A) + p_{AB} \ln(p_{AB}/y'_A y''_B) \\ & + p_{BA} \ln(p_{BA}/y'_B y''_A) + p_{BB} \ln(p_{BB}/y'_B y''_B)] \\ & + RT[b y'_A \ln y'_A + b y'_B \ln y'_B + c y''_A \ln y''_A + c y''_B \ln y''_B]. \end{aligned} \quad (22.58)$$

For a reciprocal solution, i.e., a quaternary phase where each sublattice dissolves only two of the four elements, we would obtain per mole of formula unit

$$\begin{aligned} G_m = & x_A^\circ G_A + x_B^\circ G_B + x_C^\circ G_C + x_D^\circ G_D + p_{AC} \Delta^\circ G_{A_bC_c} \\ & + p_{AD} \Delta^\circ G_{A_bD_c} + p_{BC} \Delta^\circ G_{B_bC_c} + p_{BD} \Delta^\circ G_{B_bD_c} \\ & + RTbz' \cdot [p_{AC} \ln(p_{AC}/y'_A y''_C) + p_{AD} \ln(p_{AD}/y'_A y''_D) \\ & + p_{BC} \ln(p_{BC}/y'_B y''_C) + p_{BD} \ln(p_{BD}/y'_B y''_D)] \\ & + RT[b y'_A \ln y'_A + b y'_B \ln y'_B + c y''_C \ln y''_C + c y''_D \ln y''_D]. \end{aligned} \quad (22.59)$$

This would be a case of short-range order at complete long-range order. Again we have four p_{ij} related by $\sum p_{ij} = 1$ but their relation to the composition now yields two independent equations instead of one given by Eqs (22.6) or (22.7). In this case it is most convenient to use the y_i variables which are here fixed by the composition,

$$p_{AC} + p_{AD} = y'_A \quad (22.60)$$

$$p_{AC} + p_{BC} = y''_C. \quad (22.61)$$

Consequently, now there is just one independent internal variable, one of the four p_{ij} , instead of two, and it represents short-range order because the long-range order is here perfect. It may be convenient to define the short-range order in a way similar to Eq. (22.30). However, since each kind of bond can now go in one direction, only, we shall define

$$\text{s.r.o.} = p_{AC} - y'_A y''_C. \quad (22.62)$$

We shall denote this variable by s and by combination with the relations of p_{ij} to the

composition we find

$$p_{AC} = y'_A y''_C + s; \quad dp_{AC} = ds \quad (22.63)$$

$$p_{AD} = y'_A y''_D - s; \quad dp_{AD} = -ds \quad (22.64)$$

$$p_{BC} = y'_B y''_C - s; \quad dp_{BC} = -ds \quad (22.65)$$

$$p_{BD} = y'_B y''_D + s; \quad dp_{BD} = ds. \quad (22.66)$$

The equilibrium value of s is obtained from

$$\begin{aligned} (\partial G_m / \partial s)_{y'_A, y''_C} &= \Delta^\circ G_{A_b C_c} - \Delta^\circ G_{A_b D_c} - \Delta^\circ G_{B_b C_c} + \Delta^\circ G_{B_b D_c} \\ &\quad + RTbz' \cdot [1 + \ln(p_{AC}/y'_A y''_C) - 1 - \ln(p_{AD}/y'_A y''_D) \\ &\quad - 1 - \ln(p_{BC}/y'_B y''_C) + 1 + \ln(p_{BD}/y'_B y''_D)] \\ &= \Delta^\circ G_{AC+BD} + RTbz' \cdot \ln(p_{AC} p_{BD} / p_{AD} p_{BC}) = 0 \end{aligned} \quad (22.67)$$

$$\frac{p_{AC} p_{BD}}{p_{AD} p_{BC}} = \exp(-\Delta^\circ G_{AC+BD} / RTbz'). \quad (22.68)$$

This is a result characteristic of the quasi-chemical approach. The notation $\Delta^\circ G_{AC+BD}$ was introduced in Eq. (21.20) and then used in Section 21.5.

Exercise 22.8

Consider a solution phase defined by the formula $(A, B)_b(C, D)_c$ with $N_A = N_B$ and $N_C = N_D$. Apply the compound energy formalism and estimate how much the short-range order decreases the Gibbs energy if changes in the configurational entropy are neglected. Suppose $z' = 8$, $b = c = 0.5$ and $\Delta^\circ G_{AC+BD} = 8RT$.

Hint

According to the instruction, we should study: $p_{AC}\Delta^\circ G_{A_b C_c} + p_{AD}\Delta^\circ G_{A_b D_c} + p_{BC}\Delta^\circ G_{B_b C_c} + p_{BD}\Delta^\circ G_{B_b D_c}$ which can be changed to $y'_A y''_C \Delta^\circ G_{A_b C_c} + y'_A y''_D \Delta^\circ G_{A_b D_c} + y'_B y''_C \Delta^\circ G_{B_b C_c} + y'_B y''_D \Delta^\circ G_{B_b D_c} + s(\Delta^\circ G_{A_b C_c} - \Delta^\circ G_{A_b D_c} + \Delta^\circ G_{B_b C_c} - \Delta^\circ G_{B_b D_c})$. All site fractions are fixed ($= 0.5$). The change is thus given by the last term and it can be written as $s \cdot \Delta^\circ G_{AC+BD}$.

Solution

$\Delta^\circ G_{AC+BD} / RTbz' = 2$. Inserting Eqs (22.63) to (22.66) in the quasi-chemical equation, Eq. (22.68), yields $(1/4 + s)^2 / (1/4 - s)^2 = \exp(-2)$; $s = -0.1155$; $s \cdot \Delta^\circ G_{AC+BD} = -0.1155 \cdot 8RT = -0.92RT$. This is an appreciable part of $\Delta^\circ G_{AC+BD}$.

22.8 Interstitial ordering

Interstitial solutions can also show ordering. Two effects may be recognized. The first effect is a tendency for the interstitial atoms to avoid occupying sites which are nearest

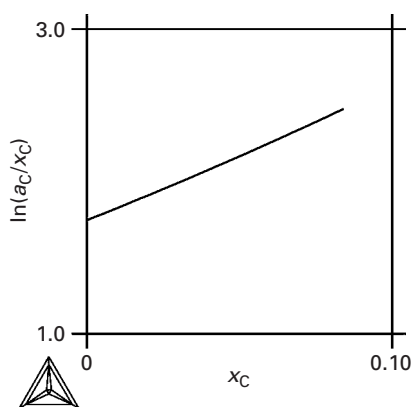


Figure 22.1 The activity coefficient of C in fcc-Fe at 1400 K as function of the C content.

neighbours to each other. In the extreme case, all the sites which are nearest neighbours to an interstitial atom will be excluded from occupancy. ‘Excluded-sites’ models have been developed for this case. As an example, we may consider fcc-Fe where the interstitial sites form their own fcc sublattice. Each interstitial site is surrounded by 12 nearest-neighbour interstitial sites. According to the model, each interstitial atom in a dilute solution excludes its own site and 12 neighbouring interstitial sites from being occupied by other atoms. For dilute solutions the lattice would thus behave as if it should be completely filled at a value of $y_C = 1/13$ where C represents the interstitial component. The partial entropy of C would thus be

$$S_C = -R \ln \frac{y_C}{1 - 13y_C} \cong -R(\ln y_C + 13y_C). \quad (22.69)$$

In the ideal case, i.e. neglecting all other interactions, this model will predict a very strong positive deviation from Henry’s law. Experimental information on the carbon activity in fcc-Fe–C does show a strong positive deviation but not enough to satisfy the excluded-sites model. This is demonstrated by Fig. 22.1 where $\ln(a_C/x_C)$ has been plotted versus x_C . The slope is about 8 but should have been 14 according to the excluded-sites model because $y_C/(1 - 13y_C)$ is equal to $x_C/(1 - 14x_C)$ when $a = 1$ for both sublattices.

In order to obtain better agreement for carbon in fcc-Fe one has to relax the condition of absolute exclusion to one in which a neighbouring site can be occupied at the expense of a certain energy v . Such models are sometimes called statistical models. A treatment can also be based upon the quasi-chemical approach which will formally apply to interactions between atoms in the interstitial sublattice just as well as to the ordinary lattice in a substitutional solution. By substituting y for x in Eq. (22.42) we obtain

$$G_m = y_C^\circ G_{M_bC} + y_{Va}^\circ G_{M_bVa} + RT(y_C \ln y_C + y_{Va} \ln y_{Va}) \\ + v z N^A y_C y_{Va} [1 - y_C y_{Va} v/kT - (2/3)y_C y_{Va}(y_C - y_{Va})^2(v/kT)^2]. \quad (22.70)$$

With the rule of calculation given for G_C in Eq. (21.12) we obtain, for composition-independent v ,

$$\begin{aligned}
 RT \ln a_C = G_C - {}^0G_C = {}^0G_{M_bC} - b {}^0G_M - {}^0G_C^\alpha \\
 + RT \ln[y_C/(1 - y_C)] + v z N^A (y_{Va} - y_C)[1 - 2y_C y_{Va} v/kT \\
 + 8y_C y_{Va}(y_C y_{Va} - 1/6)(v/kT)^2].
 \end{aligned}
 \quad (22.71)$$

For small y_C the last term can be approximated by $v z N^A \{1 - 2y_C[1 + v/kT + (2/3)(v/kT)^2]\}$. By comparing with the power series treatment of a random interstitial solution in Section 21.2 we see that $v z N^A$ corresponds to ${}^0L_{CVa}$ if v is independent of composition. For $v/kT = 0$ we thus have complete agreement with that expression. For small v/kT there will be a difference of $-2y_C v z N^A \cdot [v/kT + (2/3)(v/kT)^2]$. This difference is initially negative for positive as well as negative v which is natural because the random mixing model always results in higher Gibbs energies than a model which allows the arrangement of the atoms to be adjusted to minimize the Gibbs energy. For a given value of v , or 0L , the quasi-chemical model will thus predict a slightly weaker deviation from Henry's law.

At high enough values of the interstitial content and of $-v/kT$ there will be long-range order, an example being the γ' phase in the Fe–N system. It has a nitrogen content corresponding to Fe_4N and the Fe atoms have the same arrangement as in fcc-Fe. It may be regarded as an ordered interstitial solution of N in fcc-Fe. It is interesting to note that no N atoms occupy nearest neighbour sites. This phase is thus a perfect example of the excluded-sites principle. The high N content, compared to the limiting value of $y_C = 1/13$ given previously, is due to the fact that each excluded site is now nearest neighbour to not only one interstitial atom but four which does not happen in a dilute solution.

In the Fe–N system there is a two-phase field between the disordered solution of N in fcc-Fe and the ordered Fe_4N phase. That reveals that there is a first-order ordering transition. The situation is different for interstitial solutions in hcp metals where one can sometimes observe that the change from the disordered, dilute solution to an ordered M_2I phase is gradual, i.e., without any kind of transition. The explanation is that the distance between the interstitial sites is shortest in the hexagonal c direction. They are thus arranged in strings and, as mentioned in Section 22.4, the number of nearest neighbors is thus $z = 2$. This is a rare case where the quasi-chemical model is not limited to dilute solutions but applies all the way to a maximum content of M_2I .

Exercise 22.9

What value of ${}^0L_{CVa}$ in the random mixing model or v/kT in the quasi-chemical model is required in order to explain the slope of 8 in the plot of $\ln(a_C/x_C)$ versus x_C for carbon in fcc-Fe?

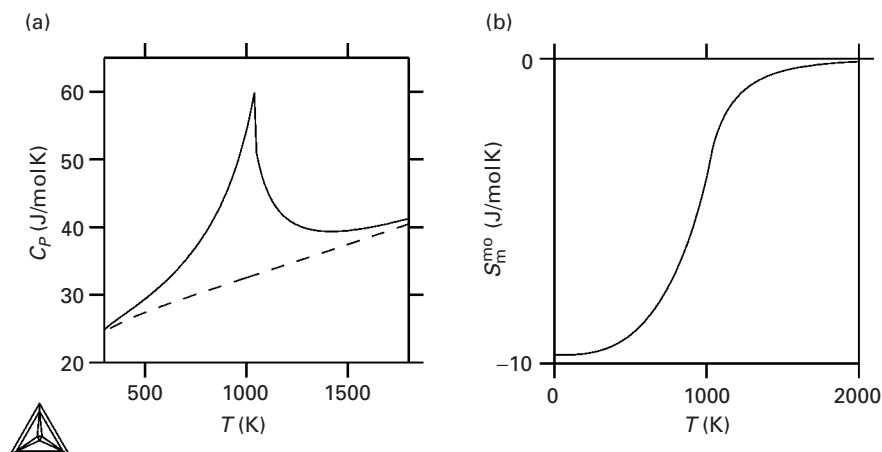


Figure 22.2 The effect of magnetic ordering on the heat capacity and entropy of bcc-Fe.

Hint

The entropy contribution to G_C in Eq. (21.11) gives $\ln[y_C/(1 - y_C)] = \ln[x_C/(1 - 2x_C)] \cong \ln x_C + 2x_C$. Remember $z = 12$.

Solution

The random mixing model for an interstitial solution gives $RT \ln(a_C/x_C) \cong \text{const} + 2RTx_C - {}^0L \cdot 2y_C \cong \text{const} + (2RT - 2{}^0L)x_C$, yielding a slope of $2RT - 2{}^0L = 8RT$; ${}^0L/RT = -3$. The quasi-chemical model gives, for small y_C , $RT \ln(a_C/x_C) \cong \text{const} + 2RTx_C - zvN^A(-2y_C)[1 + (v/kT) + (2/3)(v/kT)^2 + \dots] \cong \text{const} + \{2RT - 2zvN^A[1 + (v/kT) + (2/3)(v/kT)^2]\} \cdot x_C$, yielding a slope of $2RT - 2zvN^A[1 + (v/kT) + (2/3)(v/kT)^2] = 8RT$; $-24(v/kT)[1 + (v/kT) + (2/3)(v/kT)^2] = 8 - 2$; $v/kT = -0.339$, corresponding to ${}^0L/RT = zv/kT = 12 \cdot (-0.342) = -4.07$.

22.9 Composition dependence of physical effects

In Section 18.7 it was suggested that one should sometimes model the effect of a special physical phenomenon separately. A typical example would be the effect of magnetic ordering in a ferromagnetic material. That effect will now be used to demonstrate how the composition dependence of such an effect can be taken into account.

Figure 22.2 shows (a) the magnetic contribution to the heat capacity and (b) the decrease of the magnetic entropy from a disordered state of bcc-Fe at very high temperature. The peak temperature for the heat capacity and the inflexion point for the magnetic entropy represent the Curie temperature, T_C , i.e. the transition point between

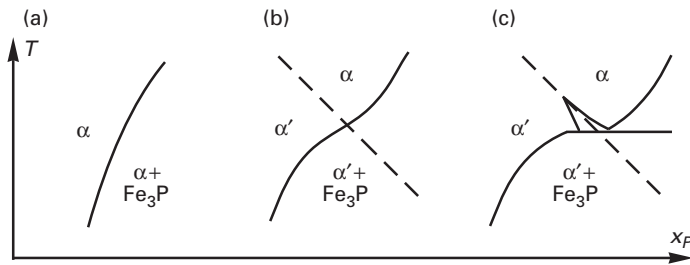


Figure 22.3 Solution to Exercise 22.11.

the high-temperature paramagnetic state which has only magnetic short-range order and the low-temperature state with long-range magnetic order.

A very simple model of magnetic ordering was presented in Section 19.6. It takes into account only long-range order. It would be possible also to include short-range order with a quasi-chemical type of approach. However, it would still be a very primitive model. For reasonably accurate modelling it seems necessary to rely on experimental measurements and some simple assumption of the dependence on composition. There have been several suggestions of mathematical expressions to be used. The first was made by Inden [48] and it is still the most widely used model. He described the heat capacity with a logarithmic expression but in order to derive an analytical expression for the contribution to the Gibbs energy it was later found necessary to use a series expansion. With truncation after the third term it yields the following expressions above and below the Curie temperature.

$$G_m^{\text{mo}} = -RT \ln(\beta + 1) \left(\frac{\tau^{-5}}{10} + \frac{\tau^{-15}}{315} + \frac{\tau^{-25}}{1500} \right) / \left[\frac{518}{1125} + \frac{11692}{15975} \left(\frac{1}{p} - 1 \right) \right] \text{ for } \tau > 1 \quad (22.72)$$

$$G_m^{\text{mo}} = RT \ln(\beta + 1) \left\{ 1 - \left[\frac{79\tau^{-1}}{140p} + \frac{474}{497} \left(\frac{1}{p} - 1 \right) \left(\frac{\tau^3}{6} + \frac{\tau^9}{135} + \frac{\tau^{15}}{600} \right) \right] \right\} / \left[\frac{518}{1125} + \frac{11692}{15975} \left(\frac{1}{p} - 1 \right) \right] \text{ for } \tau < 1, \quad (22.73)$$

where p is a constant defined as the fraction of the total disordering enthalpy which is absorbed above the critical temperature. It was given as 0.28 for fcc metals and 0.40 for bcc metals. τ is a normalized temperature defined as T/T_C . The superscript 'mo' refers to magnetic order. The equations contain two material constants, the Curie temperature T_C and the Bohr magneton number β . The equations could hopefully be applied to solutions by inserting experimental or estimated expressions for their composition dependence.

The two diagrams in Fig. 22.2 were calculated from these equations after an assessment of the magnetic properties of bcc-Fe. However, experimental measurements indicate that C_P goes to infinity at T_C from both sides (see Fig. 19.4) and that result was originally modelled by Inden. It was abandoned for practical reasons when a Gibbs energy expression was derived by integration. This approximation may seem as a bad one from a theoretical point of view but for the calculation of phase equilibria it seems to have a

comparatively small effect. An exception may be the position of the tri-critical point in a binary diagram like Fig. 15.15. According to the crude treatment of the effect of C_P^p , presented in Section 15.5, the tri-critical point should approach the side of the binary system if the magnetic C_P goes to infinity at T_C .

In order to extend the magnetic model to alloys it is necessary to make the model parameters composition-dependent. So far, investigators have studied the effect of composition-dependent β and T_C but kept the p parameter constant.

Exercise 22.10

Use the C_P curve in Fig. 22.2(a) for an estimate of p , the fraction of the total disordering enthalpy which has not yet been absorbed on heating to T_C because of the remaining short-range order.

Hint

Divide the area between the solid and dashed curves into two, one above and one below T_C , and make a crude graphical integration of these two areas.

Solution

The two areas are proportional to the two parts of ΔH . Thus $p = A_1/(A_1 + A_2) = 1/(1 + A_2/A_1) \cong 1/(1 + 1.5) = 0.4$.

Exercise 22.11

Show schematically how the solubility of phosphorus (P) in bcc-Fe can be expected to vary with temperature around the intersection with the magnetic transition line, the Curie line. Consider three different conditions: (a) no magnetic effect; (b) a magnetic effect but no separation into regions of different composition; (c) a magnetic effect and separation into regions of different composition, resulting in a miscibility gap with a tri-critical point.

Hint

Consult Section 15.5 and, in particular, Fig. 15.16 and Exercise 15.7.

Solution

(a) Without a magnetic effect we expect that $\ln(x_p)$ should vary linearly with $1/T$ (see Exercise 11.4). (b). The slope should be smaller where C_P is larger due to the magnetic effect. (c) See Fig. 22.3.